THE MICRO-DYNAMICS OF SKILL MIX CHANGE IN A DUAL LABOR MARKET:

THE SPANISH MANUFACTURING EXPERIENCE

by

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Abstract

As in many other developed countries, the share of skilled workers in Spain's labor force dramatically increased during the 1990s. This paper decomposes the aggregate skill mix change by a set of key firm characteristics and in the context of Spain's dual labor market. We find that continuing firms were the major drivers of skill mix growth and that expanding firms in particular increased their ratio of skilled workers. Net entry played a smaller but positive role due to higher-skilled entrants and lower-skilled exiters. Finally, we find that although firms with higher concentrations of temporary workers make bigger employment changes overall, firms' low-skilled employment is more strongly pro-cyclical than is high skilled employment.

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I. Introduction

Over the past several decades both the share of skilled workers and their wage premiums have increased in most industrialized countries.[1] For example, using data on individual Spanish workers, Bover, Bentolila and Arellano (2000) find that Spain had a measurable increase in earnings inequality during the 1980s, largely because of rapid wage growth at the top of the earnings distribution. Similarly, Torres (2002) showed a growth in the skilled/unskilled earnings differential beginning in 1984 and extending into the early 1990s. Similar patterns have been found by authors looking at other OECD countries' labor markets.

As economists have attempted to explain the increasing demand and wage gap, they have developed two main theories. The first is that changes in product demand, possibly due to increased participation in international trade, led to the expansion of skill-intensive products and industries (Zhu (2005)) and therefore to an increase in the demand for skilled workers. Alternative explanations tie changes in the demand for skilled workers to the introduction of skill-biased technologies across a broad spectrum of industries.[2] When technology adoption is pervasive across industries if skilled labor is a complement to the new equipment or processes, then the relative demand for skilled workers should increase. At this point the weight of the empirical evidence tilts towards the skill-biased technological change hypothesis but data limitations make conclusions difficult to reach.

While many studies have focused on unraveling the underlying determinants of the increasing gap, little exploration has been done on the micro-patterns of the of the aggregate

[1] For example, see Zhu (2005), Castillo and Jimeno (1997), Castillo (1996), Bover, Bentolila & Arellano (2000) in Spain, and Berman, Bound & Griliches (1994), Davis & Haltiwanger (1994) in the U.S.

[2] See, for instance, Bound and Johnson (1992), Davis and Haltiwanger (1991), Sachs and Shatz (1994), Berndt, Morrison and Rosenblum (1992), Berman, Bound and Griliches (1994), Autor, Katz and Krueger

skill-mix change. This paper addresses that gap. We apply a decomposition methodology similar to Baily, Hulten and Campbell (1992) and Foster, Haltiwanger and Krizan (1998) (FHK from here forward) to longitudinal micro-data from Spain to characterize the reallocation of skilled labor across firm types. We decompose how each firm type contributes to the aggregate change, paying particular attention to the dual nature of the Spanish labor market.

A special characteristic of the Spanish labor market is that it is effectively two markets: one for permanent employees and another for temporary workers. We highlight this dual labor market in a set of sub-decompositions, including some that examine differences across labor markets during a business cycle (i.e., one recession and one recovery period). Throughout the paper, we also attempt to compare our results to those obtained by Dunne, Haltiwanger and Troske (1996) using U.S. data (hereafter referred to alternatively as DHT or simply, "the U.S. results").

Our results from the main decomposition indicate that the changes in aggregate skill-mix primarily come from the within component. When we follow the decomposition results over a single business cycle (making generalizations difficult), we find that unlike in the U.S., they seem to be procyclical. That is, the bigger increase in skill mix occurred during Spain's recovery period while in the U.S., the bigger increases occurred during recessions.

The sub-decompositions by type of firm and time period show that during the recessionary period (1990-94) the firms with relatively high percentages of temporary contracts account for slightly more of the change in skill-mix. In the recovery period (1994-98) the firms with a higher shares of permanent contracts account for most of the change.

(1997), DiNardo and Pischke (1997), Doms, Dunne and Troske (1997), Luque and Miranda (2000), Entorf, Gollac and Krazmarz (1999).

When we drill down and examine the cyclicality of the changes in the percentages of skilled and less-skilled workers of 'high-permanent' and 'low-permanent' firms, the 'low-permanent' firms are on average more sensitive to the business cycle and may be fulfilling their role of lending more flexibility to the Spanish labor market.

In sum, the paper makes three contributions to the empirical literature on aggregate skill mix change. First, it explores the micro-dynamics of changes in aggregate skill mix in the Spanish labor market during the last decade and contrasts the findings with those observed in the U.S.. Second, it explores the role of the Spanish dual labor market in aggregate skill-mix changes. Thirdly, it examines their behavior across a business cycle.

The structure of the paper is as follows. Section II describes the estimation methodology and Section III describes the data that we use. Section IV first presents the results from the main decomposition and explores some of the patterns in the skill-mix changes we observe. We also present results from sub-decompositions that help illustrate: i) what types of firms are mainly responsible for the changes in aggregate skill mix, ii) how these changes seem to occur, and iii) the cyclical pattern of these changes. We conclude in Section V.

II. Estimation Methodology

A) Basic Decomposition

Our first step is to look at whether Spain's skill mix change was primarily a within-firm, between-firm or net entry phenomenon, and to see if the shares change during a business cycle. We decompose aggregate skill mix changes into a within-firm effect that reflects an increase in a firm's skilled-labor share, a between-firm effect that captures the reallocation of the employment from continuing low-skill to continuing high-skill firms, a covariance term and finally, a net entry term that is positive when entering high-skill firms displace

lower-skill exiting firms.

As mentioned in the introduction, the decomposition follows that used in DHT and FHK, which is itself a modified version of that used by Baily, Hulten, and Campbell (1992). As shown in FHK, alternative decomposition methodologies can impact the results significantly, so we use a version that we believe has the clearest interpretation. Defining M_t as an index of industry skill mix, m_{it} as an index of firm-level skill mix, and s_{it} as firm *i*'s employment share in the industry, all at time t, our base decomposition takes the following form:

$$\begin{aligned}\Delta M_t \quad = \quad & \sum_{i\in c} s_{it-1}\Delta m_{it} + \sum_{i\in c}(m_{it-1} - M_{t-1})\Delta s_{it} + \sum_{i\in c}\Delta s_{it}\Delta m_{it} \\ & \sum_{i\in B} s_{it}(m_{it} - M_{t-1}) - \sum_{i\in D} s_{it-1}\ (m_{it-1} - M_{t-1})\end{aligned} \qquad \textbf{(1)}$$

The first term of the decomposition represents a within-firm component (for continuing firms) based on firm-level changes of skill mix between *t-1* and *t* for continuing firms. The changes are weighted by the firms' initial employment shares. The second term represents the between-firm component, reflecting changes in the employment shares of continuing firms. The changes are weighted by the deviation of initial firm skill mix from the initial sector skill-mix index. This implies that an increase in a firm's share contributes positively to the between-firm component (and thus, aggregate skill mix) only if the firm has a higher skill mix than the initial sectoral average. The third term represents a cross (i.e., covariance-type) term. The last two terms represent the contribution of entering and exiting firms, respectively. Together they measure the effects of net entry.

B) Sub-decomposition

As mentioned earlier, one of the objectives of this paper is to explore the role of firm heterogeneity in aggregate skill-mix change. A particularly relevant dimension to examine in this context is the role of the 'dual' Spanish labor market in the aggregate skill-mix

change. The term 'dual' refers to the two types of employment contracts that have characterized the Spanish labor market since the mid 1980s: on one hand, permanent contracts with strong employment protection legislation and bargaining power through labor unions, and on the other hand, temporary contracts, which lack employment protection and bargaining power, and have much higher turnover rates and generally lower salaries.[3]

The dual nature of the labor market in Spain gives rise to an array of interesting questions: How are the changes in skill-labor shares being accomplished? That is, what types of firms are accounting for most of the change in the aggregate skill mix, those mostly hiring temporary or permanent-contract labor? Do the percentage changes in skilled and less-skilled workers come about due to changes in the shares of permanent or temporary-contract workers? And, how are these dynamics affected by the business cycle?

To address these questions, we perform a sub-decomposition of aggregate skill mix changes and some related tabulations in which we classify firms according to whether their workforce has a relatively high or low share of workers with permanent labor contracts ('high-permanent' or 'low-permanent' firms respectively). These exercises could have implications for debates on Spanish (and other European countries') labor market liberalization if it identifies and quantifies the relative contribution of different types of employers to aggregate labor skill mix.

Our sub-decomposition takes its cue from Bernard and Jensen (1999). Specifically, our sub-decomposition takes the following form:

$$\begin{aligned}\Delta M_t \;=\; & \sum_{g=1}^{G}\sum_{i\in c} s_{it-1}\Delta m_{it} + \sum_{g=1}^{G}\sum_{i\in c}(m_{it-1}-M_{t-1})\Delta s_{it} + \sum_{g=1}^{G}\sum_{i\in c}\Delta s_{it}\Delta m_{it} \\ & \sum_{g=1}^{G}\sum_{i\in N} s_{it}(m_{it}-M_{t-1}) - \sum_{g=1}^{G}\sum_{i\in E} s_{it-1}(m_{it-1}-M_{t-1})\end{aligned} \qquad \textbf{(2)}$$

[3] Dolado et al. (2002), Antolin (1999) and Bentolila et al. (1994).

That is, we divide each component of our base decomposition into two groups, one for firms that employ mostly permanent workers, the other for firms with high shares of temporary workers.

III. Data

A) General Characteristics of the ESEE:

The data come from the Encuesta Sobre Estrategias Empresariales (ESEE), a firm-level survey conducted by the Fundación SEPI. The ESEE is an annual survey sent to a panel of Spanish manufacturing firms, particularly large firms, and includes a representative sample of firm births and deaths for each year. Thus, in the context of this paper when we say 'aggregate' change in skill mix, we will be referring to a change in the Spanish manufacturing sector. The survey is designed to change as industry composition evolves and was designed, in part, as a research tool.

Our data cover the period from 1990-1998 and each year's file has on average over 1500 firms. The reference population for the ESEE is manufacturing businesses with 10 or more employees. Firms were selected according to a selective sampling scheme. All firms with more than 200 employees (large firms) were asked to participate, and approximately 70% of the large firms respond in a given year. Firms employing 200 or less employees were chosen according to a within-industry random sampling scheme.

The ESEE is an unbalanced panel that attempts to capture the representativeness of the industry sector in Spain. Thus, aside from making every effort to maintain the continuing firms in the sample, the survey also strives to capture the entry and exit of manufacturing firms over the sample period. Newly created firms are selected and mailed surveys using the original selection criteria. Firm exits are recorded each year and can be considered a sample representing the population of firms leaving the market over the period.

The sample's representativeness has been well documented by a number of authors. For example, Fariñas and Jaumandreu (1999) performed a series of cross checks with other data sources like the EPA (Encuesta de Población Activa or Active Population Survey) and the CB (Central de Balances from the Bank of Spain) to see if the evolution of key ESSE sample variables like employment, production and prices, is representative of the underlying population. They indeed find that the growth rates of these variables from 1990 through 1999 is consistent with that derived from the other data sources. Also ESSE figures are reasonably consistent with other sources like the 'Encuesta Industrial'.[4] In addition, the representativeness of the sample is checked annually by the Fundación SEPI in its annual analysis and results publication. When comparing their figures to those offered by other sources of the aggregate population they find their "results always reasonable."[5]

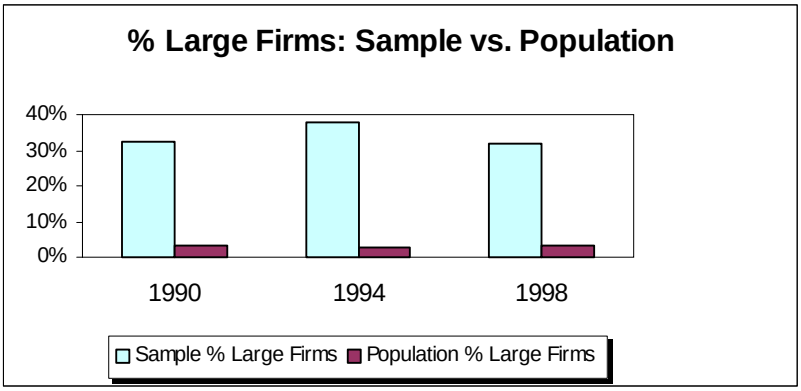
While the ESEE's aggregate properties are clearly well-established, the exercises we intend to run will lean particularly heavily on smaller businesses which are the ones most likely to enter and exit the market (Davis, Haltiwanger, and Schuh (1995)). That is, since we are particularly interested in the net entry components of our decompositions, we need to take special steps to ensure that small businesses are well-represented in our sample. Unfortunately, these are the very businesses it is most difficult to survey and track since they are less stable and have fewer resources to devote to survey response – particularly voluntary surveys.

Most statistical agencies and surveys deliberately oversample large firms since they are more likely to respond and account for a much higher proportion of aggregate activity

[4] See 'Un Panorama de la Industria Española' published by the Ministerio de Industria, Comercio y Turismo of Spain in 1992.
[5] See, for instance, 'Resultados 2000' from Fundación Empresa Pública published by the Ministerio de Ciencia y Tecnología.

(see for example the U.S. Census Bureau's Annual Survey of Manufacturers (ASM)).[6] The ESEE also oversamples large firms. For example, in as shown in Figure 1 below.

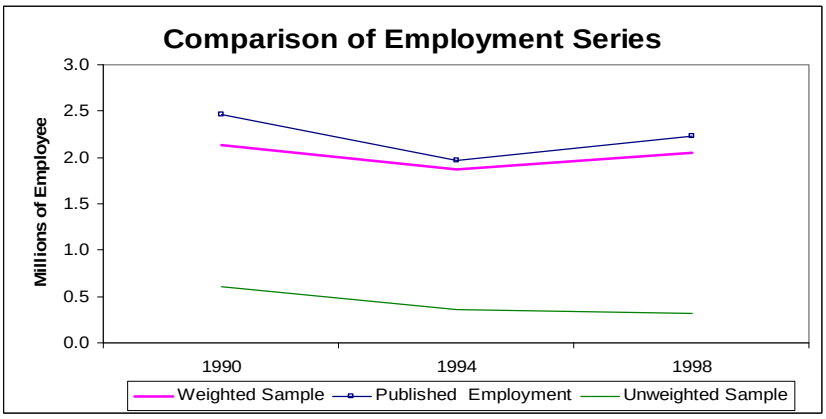
Figure 1:



Sources: Authors' computation of ESEE data and Instituto Nacional de Estadistica (www.ine.es)

To better account for the presence of the smaller firms, we apply a weighting methodology used by the U.S. Census Bureau's Annual Survey of Manufacturers (ASM) program and weight each firm by the inverse of the probability of it being in the sample.[7] The impact of this procedure can be seen in Figure 2 below:

Figure 2:



Sources: Authors' computation of ESEE data and Instituto Nacional de Estadistica (www.ine.es)

[6] http://www.census.gov/econ/overview/ma0300.html
[7] http://www.census.gov/econ/overview/ma0300.html

As Figure 2 shows, using the inverse size-based weights not only brings the aggregate totals more in-line with INE's published numbers, it also - and most importantly for our purposes - improves the sample's ability to follow the published numbers' growth pattern.

B) Special Characteristics Used in this Paper:

For this analysis we choose to focus on data from the 1990, 1994, and 1998 surveys. One of the reasons we focus on these three years is that they contain some data essential to this paper that the other years' files do not, namely information about firms' workforce skill, which is critical for our analysis. Also, while 1990 and 1998 were years of relatively high economic performance in Spain, 1994 represented the low point of a recession. Using these three years, then, allows us to look for business cycle effects in the results.

C) Definition of Skill Mix:

Following the literature (see for example, Ochsen and Welsch (2005), Acemoglu (2003), and Allen (2001)), we use educational attainment as a proxy for skill. The ESEE classifies workers into: i) 'those with a Bachelor's degree', ii) 'those with some college and/or high school diploma' and iii) 'the rest of workers'. We classify individuals with at least a Bachelor's degree as skilled labor and the workers in the other two categories as less-skilled labor. The skill mix, M_t, is the ratio between high-skill workers and all workers (including the high-skill group) in a given year.

IV. Results

A) Base Decomposition:

In this section, we present basic aggregate decompositions of the changes in skill mix focusing on the relative contributions of the within, between, covariance and *net* entry terms to aggregate skilled labor share changes. We begin by performing the decomposition for the entire eight year period (1990-1998) and then divide the sample between a recessionary (1990-94) and recovery (1994-98) period. Table 1 presents the results from the basic decomposition for all three time spans.

Within Component

The within component represent the skill upgrading that takes place within a firm. It is positive if the firm's skilled-labor share has increased. Our results indicate that this component is indeed positive and is in fact the largest of the four terms. The long difference results indicate that 67% of the total change in skill mix in manufacturing from 1990 to 1998 is accounted for by continuing firms becoming more skill intensive. In the four-year decompositions, the within effect accounts for 83 and 93 percent of the changes, indicating that in these shorter time frames almost all of the growth in skill mix came from existing firms changing the mix of their workers.

While any comparison to the U.S. results should be tempered by our recognition that DHT used production/non-production workers to define skilled/unskilled labor, our results are qualitatively similar to theirs for the U.S. They too found that the within share was the largest single source of aggregate skill mix growth.

Between Component

The between component turns out to be the smallest of the components for all difference lengths. In fact, it is essentially zero. That is, the reallocation of workers from initially low to initially high skill firms contributes little if anything to the overall change in

skill mix in Spanish manufacturing during these time frames.

The U.S. results were much different. DHT found that although it wasn't dominant, the between effect contributed up to 25% of overall skill mix long-run change between 1977 and 1982. By contrast, the between effects among Spanish manufacturing firms are either insignificant or slightly negative. A negative between term means that workers are generally moving from more skill-intensive firms to less skill-intensive firms. Or, more specifically, from firms with a skill mix above the manufacturing sector's average to firms with a skill mix below the manufacturing sector's average.

Covariance Term

Another difference between these results and those from the U.S. is that in Spain the contribution from the covariance term is relatively large and generally positive across all difference lengths. In the U.S., the covariance was small and negative in two out of the four time periods. In Spain by contrast the expanding firms increased their skill shares (and that shrinking firms' skill-shares declined) in most cases. Interestingly, the Spanish results tell us that the four-year positive covariance term occurs during the recessionary period (1990-94). That is, firms expanding their share of employment during the recession are also increasing their skill mix. Although the bigger change in aggregate skill mix does not occur during the recessionary period (as it does in the U.S.), the firms that are expanding their employment during the recessionary period are also simultaneously increasing their skill mix.

To explore the covariance term in more detail, we divide the 1990-1998 *continuing* firms into four quadrants based on their changes in employment share and skill mix. Each quadrant's contribution to the long-run aggregate skill mix changes is presented in Table 2.

We define the quadrants as follows:

Q1: Firms increasing employment share, upgrading of skill mix
Q2: Firms increasing employment share, downgrading of skill mix
Q3: Firms decreasing employment share, upgrading skill mix
Q4: Firms decreasing employment share, downgrading skill mix.

By definition quadrants 1 and 2 must have positive and negative impacts on aggregate skill mix respectively. Quadrants 3 and 4 are a bit more complex however. Firms in Q3 will contribute positively to aggregate skill unless those with above-average initial skill mixes had very large employment share reductions. This is not what we find however. Q3 firms contribute 51% of the total increase in skill mix among continuing firms. Similarly, Q4 firms – those that decreased both skill and employment share – will have a negative impact on overall aggregate skill mix unless the least productive firms dominate the category and have very large employment share losses, but this is not the case here and Q4 shows a moderate (-17%), negative effect on aggregate skill mix. These results indicate that the skill share growth among continuers came from all firms adding skilled workers to their payroll (Q1), as well as through high-skill firms eliminating low-skill jobs and low-skill firms' employment share shrinking (Q3).

Net Entry

While DHT found that net entry made a substantial contribution to aggregate skill mix change in the U.S., our results indicate that in Spain the net entry term is positive but small in the long difference. The two four year changes show that the positive long difference is the result of a positive net-entry skill mix change between 1994-1998 that compensates for a negative net entry term during the recessionary period. Since the net entry term accounts for both entering and exiting firms, it is not immediately clear if the negative sign during the 1990-1994 period is caused by entrants having a lower than average skill mix or by exiting firms having higher than average skill mixes or both. Table

3 helps sort out the two different effects by isolating the individual effects of the two net entry terms on aggregate skill mix change.

When interpreting Table 3, it is important to keep in mind that in this decomposition, the exiting term is subtracted from the entry term. That is, exiting firms contribute positively to aggregate skill mix change when they are below the average skill mix (negative term). By contrast, entrants contribute positively to aggregate skill mix change when they are relatively more skill intensive than average (positive term). With this in mind, note the interesting pattern emerging in Table 3: entrants are more skill intensive than average in the long difference because the effects of the low-skill entrants during the recession are compensated by the high-skill entrants during the recovery. By contrast, exiting firms in all time periods had lower than average skill mixes, contributing positively to the aggregate change. Table 3 also shows that in the non-recessionary period exiting firms are less skill intensive than entrants, but that during the recession, the relationship reversed and entering firms had a lower skill mix than either exiting firms or the overall average.

This same information can be perhaps more clearly seen in Table 4 where we have directly measured the percentages of skilled labor at each type of firm during each year. For example, entering firms in 1990-1994 had a particularly low share of skilled workers[8] in their labor force, only 2.2%, compared to 3.1% for firms that exited during that period and 3.3 % for continuing firms in the relevant year for the decomposition (*t-1*). In this time period, the firms that exit look more like the contemporaneous continuing firms (*(t-1)*) than do entrants. On the other hand, it is clear that the new firms in the 1994-1998 period were more skill intensive (4.2%) than both exiting firms and also previous periods entrants; in

[8] Skilled-labor shares are defined as number of skilled labor at time *t* (or *(t-1)*) divided by employment at time *t* (or *(t-1)*) for each firm type.

addition, they were more skill-intensive than the continuing firms from t-1 but less skill-intensive than the contemporaneous level of the continuers.

The other key factor in determining net entry's importance to skill-mix change is the firms' shares of overall employment – which we show in Table 5. Several points are worth noticing. First, the employment shares of exiting and entering firms vary quite a bit depending on whether the period under examination is a recession or a recovery. In the recessionary period the employment share of exiting firms (33%) is almost 4 times the one of the entering plants (9%). On the other hand, during the recovery, the employment share of both entering and exiting firms tends to be the same (around 13%). In the long difference, 1990-98 period, the employment share of exiting firms (42%) is approximately double that of entering firms (20%). Second, even over the long difference, new firms account for only 20% of all firms while exiting firms account for almost half. In sum, the small effect of net entry on aggregate skill mix seem due to the small difference between continuers and exits as well as births' small employment share and uneven relative skill level.

B) Cyclical Patterns: [9]

The U.S. results showed marked increases in the share of skilled workers during recessions that were only partially offset by mild declines during booms. Consequently, almost the entire long-run increase in aggregate skill mix occurred during economic downturns in the U.S. In Spain, this does not seem to be the case. Between 1990 and 1994 the aggregate skill mix change in Spain was 0.5% while during the 1994-1998 recovery it was 1.2%.

[9] We use the term 'cyclical' with some hesitation since our data cover only one business cycle. More data are needed before any generalizations can be made.

The proportion of skilled workers can increase either through increasing the number of skilled workers or by employing fewer less-skilled laborers. DHT spend considerable time trying to be sure that the observed counter-cyclicality in their data is not simply reflecting the firing of unskilled labor during recessions followed by their re-hiring in the recovery periods. With this in mind, Table 6 shows the percentage change in skilled and unskilled workers in both four-year periods. Interestingly, the growth in skill mix was accomplished somewhat differently during the two time periods. In the 1990-1994 period there was positive growth in the relative number of skilled workers and negative growth in the number of unskilled workers. In 1994-1998 by contrast the (larger) increase in skill mix occurred despite positive growth in the number of less-skilled workers (possibly being rehired as DHT suggest). The skilled labor share growth during the recovery occurred because there was a much larger increase in the share of skilled workers than in less-skilled workers.

C) Sub-Decompositions:

As mentioned earlier, one of the central features of Spain's labor market is its employment contract duality. This duality arose as a way to alleviate the labor market rigidity coming from the perverseness of binding employment contracts with high separation costs. In response to these contracts' negative effect on employment trends,[10] in 1984 the Spanish government created alternative types of temporary employment of workers with little or no separation costs (legal or pecuniary) upon contract termination. These flexible contracts allow larger employment responses over the course of business cycles, particularly in the downward direction, but their effect on skill mix changes is

[10] The effects of these contracts have been well documented in papers such as Bentolila and Saint-Paul (1992).

unclear. They also create a "dual" labor market with workers hired on permanent contracts, enjoying strong employment protection legislation and bargaining power through labor unions on one side, and on the other side temporary contract workers who lack employment protection and bargaining power, and have much higher turnover rates and generally lower salaries (Dolado et al. (2002)).

An interesting question to ask in the context of this paper is: how are the changes in skilled-labor shares being accomplished? That is, are the firms accounting for most of the change in the aggregate skill those that tend to hire temporary or permanent-contract labor and are there differences in how they achieve the changes? A clearly related issue is whether or not the changes in skilled and less-skilled workers come about due to changes in the shares of permanent or temporary workers. Finally, it would be interesting to know how these dynamics change over a business cycle.

Table 7 presents a sub-decomposition of aggregate skill mix by whether or not the firms are high or low permanent worker businesses. Firms are classified as "High permanent" if their average share (over the beginning and ending periods**)** of permanent-contract employees is 80 percent or more of their workforce. While any threshold is somewhat arbitrary, the 80 percent cutoff was selected because it is the median share of permanent workers for the firms in our sample. The 'Group Share' column of Table 7 shows the share of the aggregate skill mix changes from Table 3 accounted for by each firm type. The within, between, cross and net entry terms are shares of the group's contribution to the aggregate change. That is, the cross term contributed 37% of the high permanent worker group's contribution to aggregate skill mix (that is, 37% of 49%).

The first point emerging from Table 7 is that during the recession, the high-permanent firms accounted for slightly less of the aggregate skill-mix change but during the recovery the accounted for most of it. It is also clear that the skill mix changes at high-

permanent contract firms come much more through the within effect than they do for lower share firms. For these businesses, the net entry term is a much more important driver of skill mix change. That is, firms with more temporary workers also contribute to the increase in aggregate skill mix but for them it comes through the opening and closing of entire firms.

Table 8 helps us look more closely at the differences between the two groups of firms across the recession/recovery periods. It shows the percent changes in skilled and less-skilled workers by time period and firm type. Several interesting patterns emerge from this cut of the data. First, note that although both types of firms continue to hire skilled workers during the recession, there is a difference between them. High permanent-contract firms only slightly (1.5%) increase their skilled worker mix while firms with fewer permanent workers increased their skilled worker employment by about 22%. In general the changes at the firms with more temporary workers are bigger – as might be expected.

Note also that both types of firms, particularly the firms with high proportions of permanent contract workers, cut their low skilled work force at the same time. This finding may not be what one would expect a priori, but it corresponds to the evidence collected by Toharia (1998) and Malo and Toharia (1994)[11]. Given that Table 7 showed that firms with high and low shares of permanent workers accounted for roughly equal shares of aggregate skill mix change, it is interesting to note how differently they accomplished it and it would be interesting to see if the patterned held for other recessionary periods.

During the recovery period (1994-98), other patterns emerge. During this period the low permanent-contract firms made their largest employment changes – of both types of workers. It is also interesting that their percentage increase in skilled workers is higher than

[11] It seems that during the recessionary period, in particular 1993, firings of permanent-contract workers was exceptionally high due to structural adjustments in the manufacturing sector.

the percentage increase in less-skilled employees. During the two periods skilled labor's employment share increased but in different ways (though the pattern holds for both firm types). In the downturn less skilled workers lost jobs and more skilled workers were hired but during the recovery both types of workers were hired, in large numbers – but with a greater emphasis on skilled employees.

Finally, we should note that although they had lower skilled labor share growth rates, 'high-permanent' firms could account for most of the change in aggregate skill mix because they employ a much larger share of the labor force than do the 'low-permanent' firms (see Table 9).

V. CONCLUSIONS

As in other developed countries, Spanish firms' skill mix has increased during the 1990s. Using a decomposition methodology, this paper has examined the micro-dynamics of this aggregate change in skill mix over a business cycle paying special attention to a central idiosyncratic feature of the labor market in Spain.

Our decompositions showed that the largest component of the aggregate skill change is the within component. That is, the increase in aggregate skill mix mainly comes from continuing firms increasing their individual labor-skill mixes. Furthermore, the increase or upgrade in skill mix seems to be procyclical in Spain. During the 1990-1994 recession the increase in skill mix is smaller than during the recovery period (1994-98). This finding is different from the one found in the U.S., where increases in skill mix are found to be counter-cyclical. It will be interesting to explore this topic with data that covers more than one business cycle. This would help establish the pro-cyclicality or counter-cyclicality of aggregate skill-mix changes in Spain.

The results from a variety of exercises showed that during the recessionary period,

firms with a lower percentage of permanent contracts account slightly more for the increase in aggregate skill mix, but during the recovery period, the 'high permanent-contract' firms account for most of the change. Drilling a bit more into this dimension, we found that 'low permanent-contract' firms are the ones that experience the biggest percentage changes of skilled and less-skilled labor. However, due to the larger employment share of 'high permanent-contract' firms, this type of firm accounts for most of the aggregate increase in skill mix during the recovery period. Nevertheless, the evidence presented indicates that 'low-permanent-contract' firms may lend flexibility to changes in the labor market.

APPENDIX

Manufacturing Industries covered by the ESEE:

Code	Industry
1.	Metals - Metales férreos y no férreos
2.	Non-metal Mineral Products - Productos minerales no metálicos
3.	Chemical Products - Productos químicos
4.	Metallic Products - Productos metálicos
5.	Agricultural and industrial machinery - Máquinas agrícolas e industriales
6.	Office machinery, computers, etc. - Máquinas oficina, proceso datos, etc.
7.	Electronics - Material y accesorios eléctronicos
8.	Vehicles and engines - Vehículos automóviles y motores
9.	Other transportation equipment - Otro material de transporte
10.	Meat, processed food and canned goods - Carne, preparados y conservas
11.	Food products and tobacco - Productos alimenticios y tabaco
12.	Drinks - Bebidas
13.	Textiles and clothing - Textiles y vestido
14.	Leather and Shoes - Cuero, piel y calzado
15.	Wood and wood products - Madera y muebles de madera
16.	Paper, paper articles, printing - Papel, artículos de papel, impresión
17.	Plastic and rubber products - Productos de caucho y plástico
18.	Other manufactured products - Otros productos manufacturados

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Table 1: Simple Decompositions of Skill Mix Changes					
Years	Total Change in Skill Mix	Within Share	Between Share	Covariance Share	Net Entry Share
1990-1994	0.005	0.83	-0.04	0.31	-0.10
1994-1998	0.012	0.93	0.00	-0.03	0.09
1990-1998	0.016	0.67	-0.02	0.28	0.07

Table 2 **Decomposition of 1990-98 Change in Skill Mix by Long-Run Quadrant** ***(Continuing Firms)***	
Quadrant	**Contribution to Aggregate Skill Mix Change by Quadrant**
Q1 => Inc Emp Share, Upgrade Skill Mix	**0.75**
Q2 => Inc Emp Share, Downgrade Skill Mix	**-0.09**
Q3 => Inc Emp Share, Upgrade Skill Mix	**0.51**
Q4 => Inc Emp Share, Downgrade Skill Mix	**-0.17**

Table 3
Simple Breakdown of Net Entry's Contribution to Skill Mix Changes

Years	Aggregate Change In Skill Mix	Net Entry Share	Entering Firms Share	Exiting Firms Share
1990-1994	0.005	- 0.10	- 0.20	- 0.10
1994-1998	0.012	0.09	0.06	- 0.03
1990-1998	0.016	0.07	0.05	- 0.02

Table 4: Skilled Labor Shares				
	Skilled Labor Shares			
Period	**Exiting**	**Entering**	**Continuing Firms**	
	(t-1)	**(t)**	**(t-1)**	**(t)**
1990-94	0.031	0.022	0.033	0.039
1994-98	0.034	0.042	0.037	0.049
1990-98	0.033	0.037	0.034	0.053

Table 5: Employment Shares				
	Employment Shares			
Period	**Exiting**	**Entering**	**Continuing Firms**	
	(t-1)	**(t)**	**(t-1)**	**(t)**
1990-94	0.328	0.090	0.672	0.910
1994-98	0.136	0.126	0.864	0.874
1990-98	0.417	0.200	0.583	0.800

Table 6: Percentage Change in Skilled and Less-Skilled Labor		
Years	**% Change in Skilled Labor**	**% Change in Less-Skilled Labor**
1990 - 1994	7.1%	-6.6%
1994 - 1998	45.9%	9.4%

Table 7: Sub-decomposition of Aggregate Skill Mix by Firm Type						
	Firm	Group	Shares of Group Effect			
Period	Type	Share	Within	Between	Cross	Net
1990-1994	**High Perm**	0.49	1.29	-0.04	0.37	-0.61
1994-1998	**High Perm**	0.67	1.04	-0.03	-0.01	0.00
1990-1998	**High Perm**	0.77	0.69	0.04	0.31	-0.04
1990-1994	**Low Perm**	0.51	0.39	-0.02	0.25	0.39
1994-1998	**Low Perm**	0.33	0.70	0.06	-0.03	0.30
1990-1998	**Low Perm**	0.23	0.61	-0.22	0.17	0.43

Table 8 Changes in Skilled Labor by Firm Type: 1990-1994 and 1994-98				
	1990 – 1994		1994 – 1998	
Firm Type	% Change in Skilled Labor	% Change in Less-Skilled Labor	% Change in Skilled Labor	% Change in Less-Skilled Labor
High Permanent	1.5%	- 8.1%	36.0%	1.8%
Low Permanent	21.8%	- 4.3%	75.1%	22.0%

Table 9: Shares of Employment by Firm Type and Time Period				
		PERIOD		
FIRM TYPE		1990 – 1994	1994 – 1998	1990 – 1998
High-Permanent	Employment share in beginning year	0.59	0.63	0.62
	Employment share in end year	0.58	0.59	0.61
Low-Permanent	Employment share in beginning year	0.41	0.37	0.38
	Employment share in end year	0.42	0.41	0.39